

Early recruitment of boreal forest trees in hybrid poplar plantations of different densities on mine waste rock slopes

Hugo Bouchard^{a,b}, Marie Guittonny^{a,*}, Suzanne Brais^b

^a Research Institute on Mines and Environment, Université du Québec en Abitibi-Témiscamingue, 445 boul. de l'Université, Rouyn-Noranda, Québec J9X 5E4, Canada

^b Forest Research Institute, Université du Québec en Abitibi-Témiscamingue, 445 boul. de l'Université, Rouyn-Noranda, Québec J9X 5E4, Canada



ARTICLE INFO

Keywords:

Mine revegetation
Afforestation
Nurse plantation
Hydroseeded cover
Understory micro-environment
Forest succession

ABSTRACT

Mine wastes create harsh recruitment conditions for forest tree seedlings, especially waste rock piles where erodible slopes are prone to drought. Plantations using fast-growing tree species can potentially accelerate the conversion of degraded mine sites into forests through facilitation of tree recruitment, while contributing to the stability of slopes. In this study, hybrid poplars were tested as a means of achieving reclamation objectives by providing shelter for forest tree seedlings on waste rock slopes (3H:1V ratio) in the Canadian southern boreal region. Density effects of young hybrid poplars were assessed on the emergence and survival of early, mid and late successional species, naturally occurring or hand-seeded, and on the understory micro-environmental parameters in plantations of different spacings (1×1 , 2×2 , 4×4 m and control without planted trees). Results were also compared in 2×2 -m plantations with and without a hydroseeded herbaceous cover, traditionally used to control erosion in slopes. During the 2nd growing season of the plantations, seedling emergence of naturally established *Salicaceae* (*Populus* and *Salix*) species followed a quadratic pattern along the density gradient, as emergence values were higher under an intermediary density. Nonetheless, decrease in light transmission emerged as a limiting factor of seedling survival for these early-successional, shade-intolerant species by the next summer. Following a spring sowing experiment in the 3rd growing season of the plantations, emergence rates for later-successional *Picea glauca* and *Abies balsamea* seedlings increased with hybrid poplar density. During their peak emergence period, in early season, higher soil moisture content was found under denser cover. However, at the end of the third year of the plantations, only *A. balsamea* showed moderate increase in early recruitment success rates under denser tree cover. In hydroseeded plots, a competitive effect of the herbaceous cover was observed on *Salicaceae* emergence and *A. balsamea* survival. These results suggest that planting of young plantations without a hydroseeded cover may offer a more suitable solution in order to quickly provide early recruitment opportunities for later-successional seedlings in waste rock slopes. Despite this, a significant decrease in moisture content recorded during the second half of the 3rd growing season under the 1×1 -m cover, compared to the 2×2 -m, likely signalled an increasing competitive effect from hybrid poplars, which may compromise their nursing potential in the longer term. Therefore, further monitoring is imperative for a better understanding of longer-term facilitation and competition interactions between nurse trees and understory seedlings in waste rock slopes, where competition for limited resources, such as water, may be severe.

1. Introduction

Understanding the processes involved in ecosystem recovery provides useful insights for re-establishing successional trajectories towards productive and self-sustaining ecosystems (Del Moral and Walker, 2007; Walker and del Moral, 2009; Polster, 2011). There is growing scientific evidence that tree plantations can have a *catalytic effect* (Parrotta et al., 1997) on forest succession on severely degraded sites, where ecological barriers would otherwise impede recolonization

by native species (Guariguata et al., 1995; Parrotta, 1995; Carnevale and Montagnini, 2002; Boothroyd-Roberts et al., 2013). In the boreal region, vast areas of land supporting forest ecosystems are rendered unproductive because of mining activities. Recruitment is often the stage that hinders the natural regeneration of native forest trees (Young et al., 2005). The facilitation successional theory (Connell and Slatyer, 1977), as applied in restoration practice, suggests that planting of pioneer tree species able to grow on the newly exposed mine landform will assist the colonization of other species into the restored community

* Corresponding author.

E-mail addresses: hugo.bouchard@uqat.ca (H. Bouchard), marie.guittonny@uqat.ca (M. Guittonny), suzanne.brais@uqat.ca (S. Brais).

<https://doi.org/10.1016/j.foreco.2018.07.003>

Received 17 May 2018; Received in revised form 29 June 2018; Accepted 1 July 2018

Available online 01 August 2018

0378-1127/ © 2018 Elsevier B.V. All rights reserved.

(Nichols et al., 2010), and especially of later-successional species. The stress-gradient hypothesis predicts that positive interactions among plants increase with abiotic stress exposure (Bertness and Callaway, 1994), hence hinting at the potential of facilitation in reclamation efforts because of the inhospitable nature of mining substrates.

When surface-mined, all the biotic components are removed from the forested area –including tree canopy, shrub layer, soil, rootstock and seed bank (Burger and Zipper, 2002)– and replaced by accumulations of waste material. Waste rocks form one of the two main solid mine wastes along with mill tailings, and originate from rock material surrounding the ore, extracted by explosion and dumped in piles with steep slopes and flat tops (Brooks, 1990). The main forest reclamation strategy on the flat areas consists in tree planting on a respread topsoil (Drake, 1986; Kost and Vimmerstedt, 1994; Ashby, 1997; Emerson et al., 2009; Pietrzykowski, 2010; Landhäusser et al., 2012; Sloan and Jacobs, 2013). On the other hand, fast-growing herbaceous plants seeding with agronomic species is generally used on the slopes (Torbert and Burger, 1994; Aubuchon, 2010; Fields-Johnson et al., 2012), where soil erosion proves to be a significant barrier to plant establishment (Espigares et al., 2011). A ground cover of fast-growing herbaceous species allows the rapid stabilization of the soil surface and limits soil erosion on slopes (Helm, 1995). However, species commonly used (mainly grasses and legumes) often compete with tree seedlings for water, nutrients and light resources (Rizza et al., 2007; Polster, 2010; Franklin et al., 2012). Tree seedling establishment, survival and growth are thus often found to be very low on sites revegetated with herbaceous species (Andersen et al., 1989). Traditional reclamation treatments used in waste rock slopes to minimize short-term erosion may consequently hinder long-term recovery goals (Holl, 2002).

As an alternative reclamation strategy, fast-growing tree plantations could benefit the regeneration of native species that can hardly grow in open environments or in competition with a herbaceous ground cover (Carnevale and Montagnini, 2002). This nursing effect is first mediated by a *tree cover effect*. A number of mechanisms have been proposed to account for the better recruitment of native tree species under a plantation canopy, especially changes in understory microclimatic conditions (Lugo, 1997; Otsamo, 2000), reduction in competitive herbaceous species (Powers et al., 1997; Otsamo, 2000) and improvement of soil fertility through readily decomposable litter (Filcheva et al., 2000). Secondly, fast-growing trees develop an extensive root system which rapidly colonizes the available soil volume (Wilkinson, 1999; Douglas et al., 2010). Tree roots thus provide soil reinforcement that improves the stability of slopes (Abe and Ziemer, 1991), where soil erosion could adversely affect tree colonization by reducing the availability of seeds, nutrients, and water in soil (Espigares et al., 2011).

While most trees can arguably exert a facilitating role, fast-growing broadleaf species are generally regarded as better catalysts (Parrotta et al., 1997). Hybrid poplar plantations were recently observed to accelerate the colonization of native species and the restoration of forest attributes on abandoned farmlands (Boothroyd-Roberts et al., 2013). Hybrid poplar cultivars have some of the most vigorous growth among trees available for reclamation (Guy and Bateman, 1989; Casselman et al., 2006) and generally show good survival rates on mine sites (Czapowskyj, 1978; Clark Ashby, 1995; McGill et al., 2004). Some clones allocate a large proportion of their resources to roots (Larchevêque et al., 2011), which could foster the development of an extensive root system as well as a fast canopy closure to stabilize the soil and improve the understory micro-environment.

Tree spacing or density is regarded as an important factor of plantation design, potentially mediating facilitation performances in the restored community through its structuring effect on the understory micro-environment (Geldenshuys, 1997; Paquette et al., 2008; Trindade and Coelho, 2012). Denser tree covers generally offer less extreme temperatures and moisture deficiencies, but provide a more limiting light environment (Man and Lieffers, 1999). If soil moisture is known to be the main limiting factor for the germination of boreal tree species

(Greene et al., 1999), light quickly becomes limiting for the survival of shade-intolerant pioneer tree species (Karrenberg et al., 2002). Shading also hinders the development of light-demanding, weedy herbaceous species (De Keersmaecker et al., 2004). Dense weed layers not only compete for resources, but create a barrier to tree seedling establishment through leaf litter accumulation (Coates et al., 1994). The increased tree cover effect may thus maintain availability of favourable recruitment microsites and create opportunities for later-successional tree species (Boothroyd-Roberts et al., 2013), more vulnerable to desiccation than to constraints in light (Lieffers and Stadt, 1994; Landhäusser and Lieffers, 2001).

The restoration objective associated to this project was defined within a conceptual framework of community ecology (Naeem, 2006) and aimed at reestablishing native boreal tree species to restore the structure of a forest community. The facilitating role of plantations in mining conditions was examined by few studies in relation to soil re-development process (Dutta and Agrawal, 2002; Singh et al., 2004; Singh and Singh, 2006; Singh and Zeng, 2008) but remains largely unexplored regarding tree recruitment (Densmore, 2005; Frouz et al., 2015), especially on waste rock slopes. This study aimed to test the facilitation hypothesis using young hybrid poplar plantations and to understand how nurse tree spacing influences the limiting factors for the recruitment of boreal tree species in a waste rock slope (3H:1V ratio). Three hybrid poplar spacings were compared to two control treatments: bare soil without planting or seeding, and soil with planted trees and a traditional hydroseeded cover treatment. Field experiments were carried out to evaluate the effects of these plantation designs on the understory micro-environmental parameters, and on the emergence and early survival of early, mid and late successional tree species. First, we monitored soil humidity and temperature conditions, available light at ground level, leaf litter accumulation and herbaceous biomass in the plantation understories. Second, we surveyed the naturally established pioneer seedlings in each plantation designs. Third, we surveyed later-successional *Picea glauca* and *Abies balsamea* seedlings following a seed-sowing experiment. We first hypothesized that planting of hybrid poplars as nurse trees would quickly exert a structuring effect on the understory micro-environmental parameters. Secondly, it was postulated that the nurse tree cover would benefit forest tree seedling performances compared to bare soil. More specifically, we predicted better seedling emergence under denser tree cover in the young plantations because of higher soil moisture content. However, seedling mortality rates are expected to increase during canopy closure for shade-intolerant pioneer species. Thirdly, we hypothesized that the presence of a hydroseeded cover would adversely affect the emergence and survival of forest tree seedlings.

2. Materials and methods

2.1. Mine site and waste rocks

The field experiments were conducted at the Canadian Malartic mine site, located in Northwestern Quebec, Canada (48°13'N, 78°12'W). Climate is cold-temperate continental with an average annual temperature of 1.5 °C and a mean annual total precipitation of 929 mm (Government of Canada, 2015). Average length of growing season ranges between 120 and 130 days with a mean frost-free period of 97 days (Agriculture and Agri-Food Canada, 2014). The region belongs to the balsam fir-white birch bioclimatic domain in the southern portion of the boreal zone (MERN, 2003). Forest stands surrounding the mine site includes balsam fir (*Abies balsamea* (Linnaeus) Miller), black spruce (*Picea mariana* (Miller) BSP), trembling aspen (*Populus tremuloides* Michaux), white birch (*Betula papyrifera* Marshall), balsam poplar (*Populus balsamifera* Linnaeus), jack pine (*Pinus banksiana* Lambert), white spruce (*Picea glauca* (Moench) Voss) and tamarack (*Larix laricina* (Du Roi) K. Koch).

The site is an active open-pit gold mine since 2011, where 55,000

tons of ore are being processed each day. Canadian Malartic ore is a mineralized greywacke. Waste rocks have low-sulphide content (around 1% S) and contain calcite. Mean trace metal concentrations in waste rocks are below Quebec regulatory thresholds for residential land use (Government of Quebec, 2017).

2.2. Experimental setting, soil and plant material

The hybrid poplar plantations were established in May 2013 on a 50-cm overburden topsoil layer over a 3H:1V (33%) waste rock slope facing southwest and adjacent to a mature forest patch. Distance between the slope toe and the nearby forested area was slightly over 30 m. The overburden soil taken prior to ore excavation from the uppermost 30 cm (O- and A-horizons) of the partially wooded swamp above the pit was a luvic gleysol (Soil Classification Working Group, 1998). The soil contained 20% organic matter and its mineral fraction was composed of 42% clay, 27% silt and 31% sand. The overburden soil was stockpiled in 7-m-high piles (2.5:1 slope) for 36 months before being respread.

Fifteen experimental plots of 8×12 m located along the lower half of the slope and separated by 4-m-wide buffer zones were treated according to a randomized complete block design. Five treatments within three replicate blocks were applied: (1) 1×1 -m (10 000 stems/ha), (2) 2×2 -m (2 500 stems/ha) and (3) 4×4 -m (625 stems/ha) hybrid poplar spacing treatments, (4) a control treatment with an intermediary (2×2 m) poplar spacing and a traditional hydroseeded herbaceous cover, and (5) a control treatment without planting or hydroseeding (Fig. 1).

The hybrid poplar stock consisted of clonally propagated one-year-old whips (1-m long cuttings) from *Populus maximowiczii* Henry $\times$ *P. balsamifera* L. (M $\times$ B, clone number 915319), locally produced by the Ministère des Forêts, de la Faune et des Parcs du Québec (MFFP) and well adapted to the local conditions. Grass and legume species in the hydroseeded plots included the following commonly used reclamation species: *Festuca rubra* Linnaeus (15%), *Poa pratensis* Linnaeus (15%), *Pennisetum glaucum* (Linnaeus) R. Brown (12%), *Lolium perenne* Linnaeus (12%), *Avena sativa* Linnaeus (11%), *Lotus corniculatus* Linnaeus (15%), *Trifolium pratense* Linnaeus (10%), *Trifolium repens* Linnaeus (7%) and *Trifolium hybridum* Linnaeus (3%). Seeding rate was 100 kg ha^{-1} while a fertilizer (8% N, 32% P, and 16% K, di- and mono-ammonium phosphates and potassium chloride) was applied once at seeding in May 2013 at 750 kg ha^{-1} .

A. balsamea and *P. glauca* seeds were also provided by the MFFP. The seed lots received a 48-hour-priming treatment followed by a 21-day cold moist stratification treatment to promote higher germination rates (Colas and Bettez, 2014). A viability test was performed by placing 3×100 seeds of both species on wet paper in Petri dishes placed under controlled conditions at $\sim 20^\circ\text{C}$ for 20 days to assess the germination capacity of the seed lots before experimentation (Charron and Greene, 2002; Johnstone and Chapin III, 2006). The germination rates

obtained were close to those reported by the MFFP: 72 and 92% respectively for *A. balsamea* and *P. glauca*.

2.3. Understory micro-environment measurements (2014–2015)

Three 1-m^2 microsites were placed systematically within each experimental plot for micro-environment measurements during the 2014 and 2015 growing seasons. Microsites were positioned at the intersection of the diagonals of the square formed by four adjacent planted hybrid poplars in each third of the plot along the slope (Fig. 2). Continuous measures of soil water volumetric content were taken at 3-cm depth (ECH2O 5-cm probe, Decagon, calibrated for the soil type used) in the microsite located in the centre of each plot. From June to August 2014 and May to August 2015, hourly measurements were recorded to calculate weekly averages. Soil temperature at 3-cm depth (Acorn probe, Oakton Instruments) and photosynthetic photon flux density (PPFD) at ground level (Sunfleck Ceptometer, Decagon) were measured every two weeks in the three microsites of each plot during the same period. Measurements for these two parameters were taken around noon, under clear sky conditions. Aboveground herbaceous biomass was assessed during the period of maximal biomass (mid-July) in 2014 and 2015. A non-destructive point intercept method (Jonasson, 1983; Jonasson, 1988) was used to estimate the herbaceous biomass in the microsite located in the centre of each plot. A narrow rod placed perpendicularly to the soil surface was shifted along a $100 (10 \times 10\text{cm})$ intersection point grid which covered the microsite area. The number of contacts between the rod and each plant species was registered. In order to calibrate the method for biomass estimation, 12 1-m^2 quadrats (selected to include a wide range of biomass of the different species) were previously sampled in July 2014 using the same grid pattern. The vegetation within each of these quadrats was then clipped at ground level and dried in order to compute regression equations between contact points and dry biomass for the main herbaceous species. The summed number of contacts by species sampled in each experimental plot could then be used to estimate total herbaceous biomass per m^2 . Finally, total leaf litter percentage cover (from hybrid poplars and herbaceous species) was measured in early and late summer 2014 and 2015. A $25 (20 \times 20\text{cm})$ point grid was used for this measurement in the three microsites for each plot. Each intersection point where the rod touched at least one leaf on the ground was noted as a litter occurrence to estimate the percentage cover as follows: number of occurrences/total number of measurement points $\times 100$.

2.4. Natural colonization survey (2014–2015) and field sowing experiment (2015)

Naturally established pioneer woody (tree and arborescent shrub) seedlings were monitored in nine randomly positioned 1-m^2 quadrats in each plot (Fig. 2). Seedling numbers were surveyed by species during

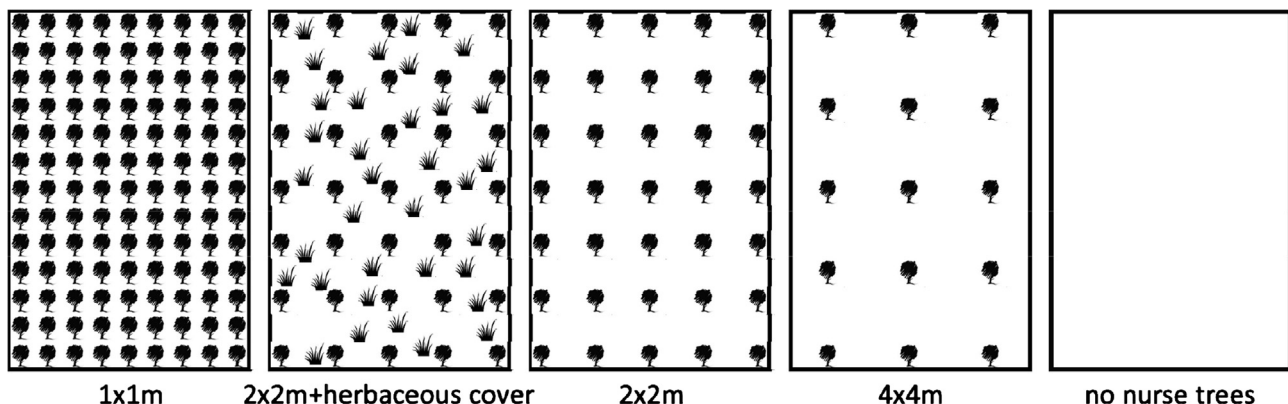


Fig. 1. Hybrid poplar plantation designs grouped in each block of the experimental layout (not drawn to scale nor randomized).

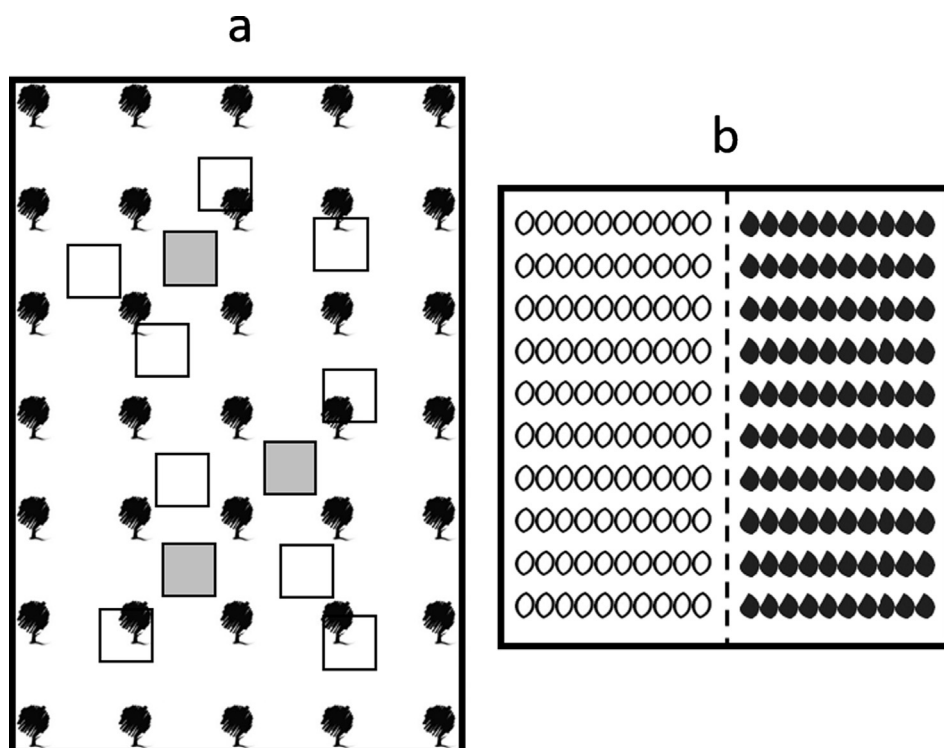


Fig. 2. Sample units in individual experimental plot. (a) Microsites for the micro-environmental measurements and the sowing experiment are symbolized by coloured squares; quadrats for the natural colonization survey, by blank squares. (b) Magnified representation of microsites, sown with *Abies balsamea* and *Picea glauca* seeds (numbers shown are different to those in the experiment).

early, mid and late summer 2014 and 2015. Data were used to quantify summer emergence (total number of new seedlings recorded during mid and late summer counts per m²), summer mortality rates (total number of dead seedlings recorded during mid and late summer counts/total number of seedlings recorded during early, mid or late summer counts) and late-season abundance (total number of seedlings recorded during the late summer count per m²) in each quadrat, for 2014 and 2015.

The sowing experiment was conducted in the three microsites already used for the micro-environment measurements in each of the 15 experimental plots. Each microsite was divided into two equal parts sown respectively with *A. balsamea* and *P. glauca* just after snowmelt in mid-May 2015. Seeds were sown by hand in tiny furrows (40 seeds × 10 rows for both species) at an interrow spacing of 10 cm and then partially covered to minimize the risk of loss by erosion on the slope (Fig. 2). A total of 1200 seeds (400 seeds × 3 microsites) per species were sown this way in each plot. Seedling numbers were surveyed for each species every four weeks from June to August 2015. Data were used to quantify summer emergence rates (total number of emergences during the growing season/number of seeds), summer mortality rates (total number of dead seedlings/total number of emerged seedlings during the season) and first-year recruitment success rates (number of surviving seedlings at the end of the growing season/number of seeds) in each microsite.

2.5. Statistical analyses

Data from the natural colonization survey and the sowing experiment were analyzed according to a randomized complete block design using generalized linear mixed models. Models were fitted with the glmer function of the lme4 package of R (Bates et al., 2015; R Core Team—version 3.2.3–2015). Maximum likelihood estimation was based on the Laplace approximation (Laplace, 1986). A multiple regression analysis was used to test whether seedling emergence, mortality and abundance: (1) increased or decreased with planted hybrid poplar density (linear relation), (2) reached a minimum/maximum value under intermediate density (quadratic relation), and (3) differed

between the 2 × 2-m-spacing treatments with and without a hydro-seeded cover. Blocks and experimental units were treated as random effects with the latter nested in the former. That is, predictive models could be summarized as follows: seedling response variable ~ linear density + quadratic density + hydroseeding + random effects (block/experimental units).

Only the *Salicaceae* family was considered for the statistical analysis of the natural colonization survey, and tests were performed for all the species combined (pooled *Populus* and *Salix* species). Seedling emergence, mortality rates and late-season abundance were analyzed on a growing-season basis. For the sowing experiment, seedling emergence, mortality and recruitment success were analyzed separately for each species.

Data from the micro-environment measurements were analyzed with linear mixed effect models using the nlme package of R (Pinheiro et al., 2015). Model parameter estimates were based on the restricted maximum likelihood method. A multiple regression analysis was used to test the effects of hybrid poplar density, as linear and quadratic terms, and hydroseeding, as a binary variable, on: soil moisture, soil temperature, available light at ground level, herbaceous biomass and total leaf litter cover. The quadratic term was removed when not significant for a better estimate of the simple term of density. The analysis was conducted for key periods of seedling responses to experimental treatments, namely: (1) the highest emergence period of *Salicaceae* species (late June to late July 2014), (2) the highest emergence period of both *P. glauca* and *A. balsamea* (mid-May to mid-June 2015) and (3) the highest mortality period for any species (mid-July to mid-August 2015). Average value of micro-environmental variables registered in microsites within each time period was used for the analysis. A prior first order autoregressive analysis was conducted for repeated measures to confirm that the effect of treatments was constant within each period studied. Data were log-transformed when necessary to meet assumptions of normality and homogeneity of variance. The logarithmic value of hybrid poplar density was also used for all regression analyses, in order to get more regular intervals between treatment levels along the density gradient.

For all analyses, we considered a significance level equal to 0.10 to

allow a broader effects-based analysis. For all models illustrated graphically, we used a Monte Carlo approach (Gelman et al., 2014) to estimate predicted values and 90% confidence intervals along the density gradient, from the median, the 5th and 95th percentiles of the posterior predictive distribution of model parameters.

Linear relationships between the micro-environmental variables and seedling emergence and mortality rates were explored using Pearson correlations for the determining periods mentioned above. A complementary analysis was conducted using the Hoeffding's D measure (Harrell Jr and Dupont, 2006) to test for dependence structures beyond linear and monotonic associations but was not retained since no additional relationships between variables were detected. For both the natural colonization survey and the sowing experiment, we used the average value of each variable during the highest emergence and mortality periods recorded. The Bonferroni correction was applied to adjust for the significance level of multiple correlation tests, dividing the alpha level by the number of tests performed for each seedling response. Since five micro-environmental parameters were considered, $p\text{-value} = 0.10/5 = 0.02$.

3. Results

3.1. Tree cover effect on the understory micro-environment (2014–2015)

Variations in the understory conditions were observed along the hybrid poplar density gradient during key periods of seedling responses in the 2nd (2014) and 3rd (2015) growing seasons of the plantations (Table 1). The effect of the hybrid poplar density on herbaceous biomass was already noticeable in July 2014, while values measured during the period of maximal biomass decreased linearly with tree density (Fig. 3). Mean dry herbaceous biomass was > 3 times lower under the $1 \times 1\text{-m}$ spacing treatment ($45 \text{ g}\cdot\text{m}^{-2}$; CI limits: 25–70) compared to the control plots without nurse trees ($150 \text{ g}\cdot\text{m}^{-2}$; CI limits: 100–225). Herbaceous biomass remained similarly related to poplar density in 2015, but was overall almost 2 times less important than that of the previous year under the developing canopies in the plantations (see appendix A for the list of inventoried herbaceous species).

Relationships between poplar density and soil moisture content followed a quadratic pattern from late June to late July 2014 (Table 1). The mean volumetric water content under an intermediary spacing treatment (26%; CI limits: 24–28) was higher than under the $1 \times 1\text{-m}$ spacing level (22%; CI limits: 20–25) or the control without nurse trees (21%; CI limits: 19–23, Fig. 3). For early season 2015, linear regression coefficients indicated a significant positive effect of poplar density on soil moisture conditions, while the highest mean volumetric water content was found under the $1 \times 1\text{-m}$ spacing treatment (36%; CI limits: 34–38) and the lowest, under the control without nurse trees (31%; CI limits: 29–33). Similarly to 2014, the moisture content under denser cover dropped compared to intermediary spacing levels in the second half of the summer. The relation between poplar density and moisture content thus switched from a linear to a quadratic pattern, with lower mean values found at both ends of the density gradient (29%; CI limits: 27–30, and 27%; CI limits: 25–28, respectively for the control and $1 \times 1\text{-m}$ treatments) from mid-July to mid-August 2015. Notwithstanding, soil volumetric water content remained higher than the wilting point measured (results not shown) for the soil material considered –with values over 18.5% for all types of cover treatment– during the whole growing season.

The influence of hybrid poplar density on available light at ground level and soil temperature was similar for the late June to late July 2014, mid-May to mid-June 2015 and mid-July to mid-August 2015 periods, showing significant linear or quadratic effects (Table 1). Mean values for these parameters were the lowest under the $1 \times 1\text{-m}$ spacing treatment throughout the survey. The $1 \times 1\text{-m}$ poplar cover intercepted on average $\sim 70\%$ of full sunlight ($600 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, CI limits: 400–810, compared to $\sim 2000 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ for full sunlight) during

measurements from late June to late July 2014 (Fig. 3). Light interception reached a maximum of $\sim 80\%$ during summer 2015, but was on average $\sim 60\%$ ($870 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, CI limits: 820–930, compared to $\sim 2180 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ for full sunlight) and $\sim 70\%$ ($610 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, CI limits: 460–770, compared to $\sim 2060 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ for full sunlight) respectively for the early and late season periods considered. Comparatively, light interception for the $2 \times 2\text{ m}$, the $4 \times 4\text{ m}$ and the control without nurse trees was roughly half that of the $1 \times 1\text{-m}$ spacing treatment throughout the survey. Likewise, soil temperature was the lowest under the $1 \times 1\text{-m}$ spacing treatment for every time period studied (Table 1, Fig. 3). Mean temperature values for the $2 \times 2\text{ m}$, the $4 \times 4\text{ m}$ and the control without nurse trees –as for available light– were generally more alike. Mean differences between the $1 \times 1\text{ m}$ plots and the other spacing levels were $< 2.5^\circ\text{C}$ from late June to late July 2014, but increased to $> 4.3^\circ\text{C}$ during both periods studied in 2015.

The density effect on total leaf litter cover (from hybrid poplars and herbaceous species) changed over the duration of the study (Table 1, Fig. 3). A quadratic pattern was observed from late June to late July 2014, reflecting a stronger contribution from hybrid poplars and herbaceous species at opposite ends of the density gradient. However, regression coefficients indicate a positive linear effect of poplar density on leaf litter cover in early season 2015, but no significant effect in late season 2015, while mean values reached the 40–50% range.

3.2. Hydroseeded cover effect on the understory micro-environment (2014–2015)

The influence of the hydroseeded cover treatment on the herbaceous biomass was significant during the period of maximal herbaceous biomass both in 2014 and 2015 (Table 1). Mean herbaceous dry biomass in hydroseeded plots ($235 \text{ g}\cdot\text{m}^{-2}$; CI limits: 140–390) was close to 3 times that of unseeded plots ($80 \text{ g}\cdot\text{m}^{-2}$; CI limits: 60–105) in 2014 and slightly more than twice that of unseeded plots ($130 \text{ g}\cdot\text{m}^{-2}$; CI limits: 110–145 vs $\sim 60 \text{ g}\cdot\text{m}^{-2}$; CI limits: 50–70) in 2015, with declining values from 2014 to 2015 (Fig. 3). The hydroseeded cover treatment also significantly affected light availability. From late June to late July 2014, $\sim 60\%$ ($790 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, CI limits: 590–990, compared to $\sim 2000 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ for full sunlight) of full sunlight was intercepted on average during measurements in hydroseeded plots compared to $< 35\%$ ($1250 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, CI limits: 1070–1430) in unseeded plots. Treatment effect on light conditions was not detected from mid-May to mid-June 2015 –before full early season regrowth– but increased later in the season, with % light transmission for the mid-July to mid-August period similar to those of 2014. Mean differences in soil temperature were not detected until mid-July to mid-August 2015 and remained marginal even during this period ($\sim 25.2^\circ\text{C}$, CI limits: 24.4–25.8, for hydroseeded plots; 26.2°C , CI limits: 25.6–26.8, for unseeded plots). The influence of the hydroseeded cover on leaf litter cover was not constant. Percentage leaf litter cover significantly differed only in early season 2015, while mean value was 36% (CI limits: 33–38) in seeded plots compared to 26% (CI limits: 25–28) for unseeded plots. Finally, soil moisture content was not impacted by the hydroseeded treatment during any of the period studied.

3.3. Natural colonization (2014–2015)

Abundance of colonizing woody (tree and arborescent shrub) species after three growing seasons was largely dominated by *Salicaceae*: *Populus* and *Salix* species accounted for 72 and 10% of total abundance respectively in the plantations. Other seedlings observed belonged to the genera (in order of decreasing abundance): *Picea* (10%), *Sorbus* (3%), *Abies* (2%), *Prunus* (1%), *Betula* (1%) and *Cornus* (1%) (see appendix B for the list of inventoried woody species). Mean abundance of seedlings in the quadrats for all genera combined was $1.3 \pm 0.2(\text{CI})\cdot\text{m}^{-2}$ at the end of summer 2014, but $1.0 \pm 0.2(\text{CI})$ seedlings per m^2 at the end of summer 2015, pointing to an overall

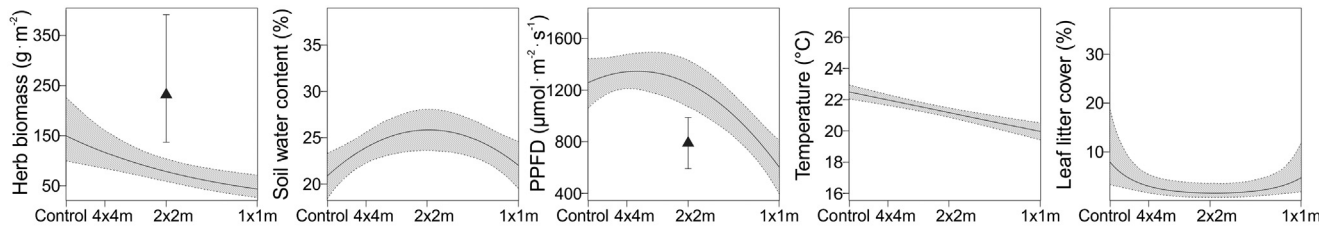
Table 1
Effect of hybrid poplar density and hydroseeded herbaceous cover treatments on herbaceous biomass, soil moisture, available light (PPFD) at ground level, soil temperature and leaf litter cover. Results are presented for the highest emergence period of *Salicaceae* (*Populus* and *Salix*) species (late June to late July 2014), the highest emergence period of *Picea glauca* and *Abies balsamea* (mid-May to mid-June 2015) and the highest mortality period for any species (mid-July to mid-August 2015) in the plantations. P-values are shown in bold when under the 10% significance threshold. N = 15 for herbaceous biomass and soil moisture; N = 45 for available light, soil temperature and leaf litter cover.

	Herbaceous biomass ^a				Soil moisture				Available light ^b				Soil temperature				Leaf litter cover ^a			
	Estimate	Std. Error	P-value		Estimate	Std. Error	P-value		Estimate	Std. Error	P-value		Estimate	Std. Error	P-value		Estimate	Std. Error	P-value	
<i>late June to late July 2014</i>																				
Intercept	4.451	0.272	< 0.001		25.728	1.539	< 0.001		1299.228	151.868	< 0.001		21.357	0.624	< 0.001		0.493	0.746	0.525	
Linear density	–0.512	0.178	0.018		1.220	0.943	0.228		–215.925	76.068	0.019		–1.051	0.414	0.029		–0.447	0.354	0.239	
Quadratic density	–	–	–		–3.047	1.269	0.040		–234.338	102.331	0.048		–	–	–		0.928	0.477	0.084	
Hydroseeding	0.991	0.437	0.049		–3.105	2.065	0.167		–515.549	166.483	0.013		0.294	0.843	0.735		0.249	0.776	0.755	
<i>mid-May to mid-June 2015</i>																				
Intercept	NA	NA	NA		33.143	0.733	< 0.001		7.395	0.083	< 0.001		21.075	0.494	< 0.001		25.555	1.394	< 0.001	
Linear density	NA	NA	NA		2.202	0.800	0.020		–0.166	0.049	0.008		–0.917	0.354	0.029		3.167	1.012	0.011	
Quadratic density	NA	NA	NA		–	–	–		–0.236	0.065	0.006		–1.696	0.476	0.006		–	–	–	
Hydroseeding	NA	NA	NA		–0.110	1.629	0.948		–0.044	0.106	0.690		–1.096	0.775	0.191		9.891	2.060	0.001	
<i>mid-July to mid-August 2015</i>																				
Intercept	62.660	5.024	< 0.001		32.333	0.989	< 0.001		1395.965	116.349	< 0.001		26.438	1.055	0.000		47.662	5.841	< 0.001	
Linear density	–12.753	5.543	0.044		–0.132	0.640	0.841		–165.920	79.613	0.067		–1.205	0.275	0.002		3.707	4.080	0.385	
Quadratic density	–	–	–		–3.253	0.861	0.004		–324.514	107.102	0.014		–0.841	0.370	0.049		–	–	–	
Hydroseeding	65.541	11.287	< 0.001		–1.104	1.400	0.451		–548.331	174.244	0.012		–1.305	0.602	0.058		9.689	8.307	0.271	

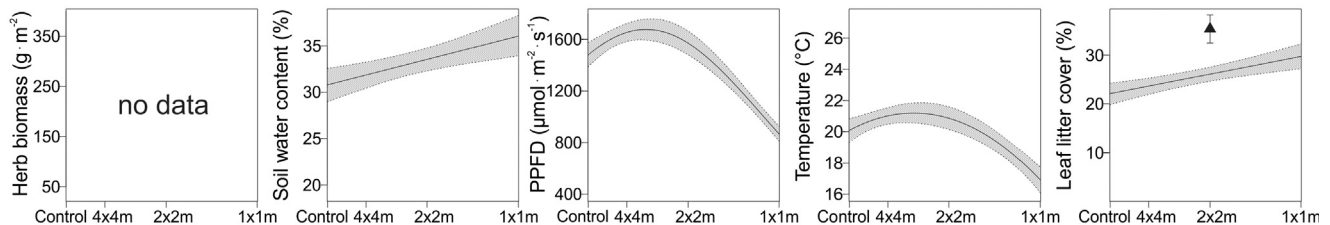
^a Log transformed for late June to late July 2014.

^b Log transformed for mid-May to mid-June 2015.

a) late June to late July 2014: highest emergence period of *Salicaceae*



b) mid-May to mid-June 2015: highest emergence period of *P. glauca* and *A. balsamea*



c) mid-July to mid-August 2015: highest mortality period for any species

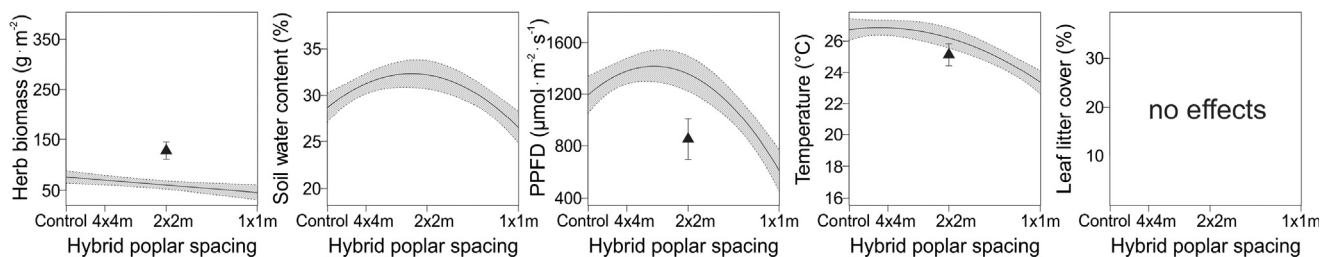


Fig. 3. Influence of hybrid poplar spacing and hydroseeded herbaceous cover treatments on the understory micro-environmental parameters in key periods of seedling responses, during (a) the 2nd (2014) and (b, c) the 3rd (2015) growing seasons of the plantations. The solid line and grey margins symbolize predicted values and 90% CIs along the hybrid poplar gradient; the triangle and bars symbolize the predicted value and the 90% CI in the 2×2 -m spacing treatment with a hydroseeded cover.

Table 2

Effect of hybrid poplar density and hydroseeded cover treatments on the emergence, mortality and late-season abundance of naturally established *Salicaceae* (*Populus* and *Salix*) seedlings. Results are presented for the 2nd and 3rd growing seasons of the plantations. P-values are shown in bold when under the 10% significance threshold. N = 135.

	Summer emergence			Summer mortality			Late-season abundance		
	Estimate	Std. Error	P-value	Estimate	Std. Error	P-value	Estimate	Std. Error	P-value
2014									
Intercept	−0.435	0.275	0.114	−2.047	0.258	< 0.001	0.338	0.436	0.437
Linear density	0.244	0.193	0.207	0.295	0.303	0.329	0.165	0.162	0.309
Quadratic density	−0.450	0.232	0.052	–	–	–	−0.385	0.202	0.056
Hydroseeding	−1.322	0.511	0.010	0.529	0.688	0.442	−1.310	0.398	0.001
2015									
Intercept	NA	NA	NA	−1.713	0.294	< 0.001	0.064	0.480	0.895
Linear density	NA	NA	NA	0.789	0.328	0.016	0.048	0.176	0.784
Quadratic density	NA	NA	NA	–	–	–	−0.466	0.206	0.024
Hydroseeding	NA	NA	NA	0.477	0.779	0.541	−1.387	0.437	0.002

emergence/mortality ratio switching from a > 1 (2.7) to a < 1 (0.4) value in the plantations. Variations in seedling emergence, mortality or abundance patterns were observed for the dominant *Salicaceae* species under both the hybrid poplar density and the hydroseeded cover treatments (Table 2). In 2014, mean seedling emergence was greater under the intermediary 2×2 -m spacing treatment ($\sim 0.7 \text{ m}^{-2}$; CI limits: 0.5–0.9) and lower under the control treatment without nurse trees ($\sim 0.3 \text{ m}^{-2}$; CI limits: 0.2–0.4), following a significant quadratic pattern along the hybrid poplar density gradient (Fig. 4). Mean emergences in 2014 were also significantly influenced by the hydroseeded treatment: nearly 3.5 times less seedlings emerged in hydroseeded plots ($< 0.2 \text{ m}^{-2}$; CI limits: 0.1–0.3) compared to unseeded plots. In

contrast, summer mortality in 2014 was not impacted by either treatment—the same applies for the 2014–2015 winter mortality (results not shown). In 2015, seedling emergence was too low to test for effects. Summer mortality rates, on the other hand, increased linearly with hybrid poplar density in 2015: mean mortality rates were 34% (CI limits: 20–49) under the 1×1 -m spacing treatment compared to 7% (CI limits: 3–13) in control plots without nurse trees. No effect of the hydroseeded cover treatment on the 2015 summer mortality was detected. At the end of the 2015 growing season, *Salicaceae* seedling abundance was greater under an intermediate level of hybrid poplar density ($p = 0.024$ for quadratic relation)—with mean values up to 1.1 m^{-2} (CI limits: 0.8–1.4) in the 2×2 -m and in plots without a

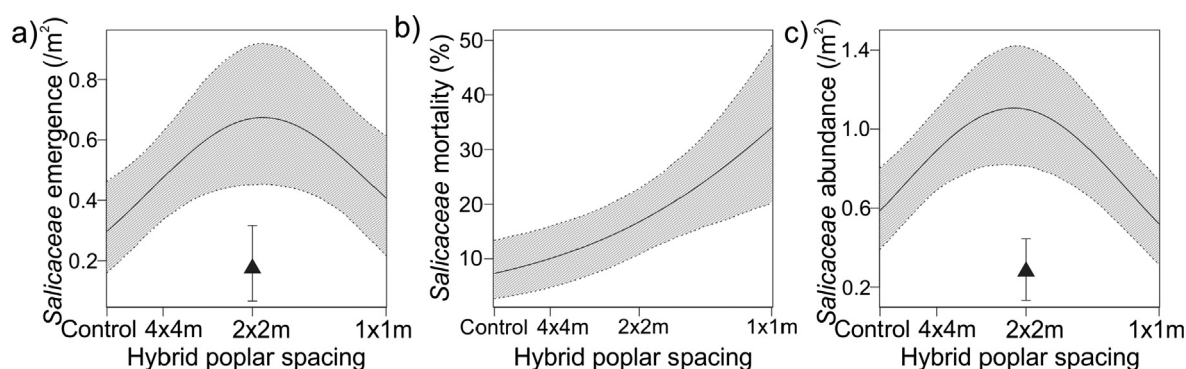


Fig. 4. Influence of hybrid poplar spacing and hydroseeded herbaceous cover treatments on seedling response of naturally established *Salicaceae* (*Populus* and *Salix*) species during the 2nd and 3rd growing seasons of the plantations. Results summarize (a) summer emergence in 2014, (b) summer mortality in 2015 and (c) late-season abundance in 2015. The solid line and grey margins symbolize predicted values and 90% CIs along the hybrid poplar gradient; the triangle and bars symbolize the predicted value and the 90% CI in the 2×2 -m spacing treatment with a hydroseeded cover.

hydroseeded cover ($p = 0.002$) –with mean value as low as $< 0.3\text{ m}^{-2}$ (CI limits: 0.1–0.4) in hydroseeded plots. In other words, differences in the pioneer seedling abundance observed after the emergence period in 2014 were still noticeable at the end of summer 2015. However, the lowest abundance values for the spacing treatment were now noted in the 1×1 -m level at the end of summer 2015 ($\sim 0.5\text{ m}^{-2}$; CI limits: 0.3–0.7).

3.4. Field sowing experiment (2015)

The hybrid poplar spacing treatment significantly influenced the summer emergence of *P. glauca* and, more marginally, of *A. balsamea*, while rates increased linearly with poplar density for both species (Table 3). Mean emergence rates were 15.7% (CI limits: 13.7–17.7) and 6.4% (CI limits: 5.1–7.9) respectively for *P. glauca* and *A. balsamea* under the 1×1 -m spacing treatment compared to 11.9% (CI limits: 10.6–13.2) and 4.4% (CI limits: 3.6–5.3) under the control treatment without nurse trees (Fig. 5). On the other hand, emergence rates remained unaffected by the presence of a hydroseeded cover for both species. After one growing season, overall mean mortality rates in microsites were 30.5 ± 3.8 (CI) and 28.6 ± 3.8 (CI)% respectively for *P. glauca* and *A. balsamea*. Mortality rates for *P. glauca* were not significantly impacted by either treatments. *A. balsamea* mortality also remained unchanged by the hybrid poplar spacing treatment, but significant differences were observed under the hydroseeded treatment ($p = 0.044$), with higher rates registered in hydroseeded plots (37%; CI limits: 27–48) compared to unseeded plots (25%; CI limits: 21–28). Overall recruitment success rates in microsites were 9.8 ± 1.0 (CI) and 4.0 ± 0.7 (CI)% respectively for *P. glauca* and *A. balsamea* first-year seedlings. The positive effect of density on recruitment success after one

growing season was significant only for *A. balsamea* ($p = 0.064$). Mean recruitment success rates for *A. balsamea* seedlings were $\sim 5.0\%$ (CI limits: 3.8–6.3) under the 1×1 -m spacing treatment compared to 3.2% (CI limits: 2.5–3.9) under the control treatment without nurse trees. On the other hand, the resulting effect of the hydroseeded treatment on recruitment success at the end of the growing season was not statistically significant for either *P. glauca* or *A. balsamea*, notwithstanding the higher summer mortality rates registered for *A. balsamea* in plots with a hydroseeded cover.

3.5. Relationships between the understory micro-environment and seedling responses

The correlation analysis did not reach statistical significance regarding relationships between the micro-environmental variables and the *Salicaceae* seedlings emergence during the 2014 emergence period (Table 4). However, *Salicaceae* seedling mortality was negatively correlated with the available light variable ($r = -0.398$, $p = 0.013$) during the main mortality period surveyed over the duration of the study, in late-season 2015. Diminution of light availability thus corresponded to increased seedling mortality rates for these species at the end of the third growing season in the plantations.

No significant relationships were identified between the micro-environmental variables and the seedling emergences of *P. glauca* and *A. balsamea* during the peak emergence period monitored in the 2015 growing season (Table 5). Strength of relationships for moisture conditions was the highest of all micro-environmental variables, both with *P. glauca* ($r = 0.533$) and *A. balsamea* ($r = 0.500$), but p-values (0.041 and 0.058) remained higher than the Bonferroni-adjusted significance threshold. Likewise, the correlation analysis did not detect any

Table 3

Effect of hybrid poplar density and hydroseeded cover on summer emergence, mortality and late-season recruitment success of *Picea glauca* and *Abies balsamea* seedlings following spring seeding in 3-year-old hybrid poplar plantations. P-values are shown in bold when under the 10% significance threshold. N = 45.

	Summer emergence			Summer mortality			Late-season recruitment success		
	Estimate	Std. Error	P-value	Estimate	Std. Error	P-value	Estimate	Std. Error	P-value
<i>Picea glauca</i>									
Intercept	−2.004	0.107	< 0.001	−1.194	0.086	< 0.001	−2.367	0.131	< 0.001
Linear density	0.116	0.057	0.041	0.149	0.092	0.108	0.029	0.067	0.660
Quadratic density	–	–	–	–	–	–	–	–	–
Hydroseeding	0.067	0.116	0.563	−0.061	0.189	0.749	0.076	0.134	0.568
<i>Abies balsamea</i>									
Intercept	−2.953	0.218	< 0.001	−1.374	0.090	< 0.001	−3.235	0.216	< 0.001
Linear density	0.161	0.093	0.085	−0.122	0.100	0.225	0.184	0.100	0.064
Quadratic density	–	–	–	–	–	–	–	–	–
Hydroseeding	−0.163	0.187	0.384	0.375	0.187	0.044	−0.339	0.211	0.108

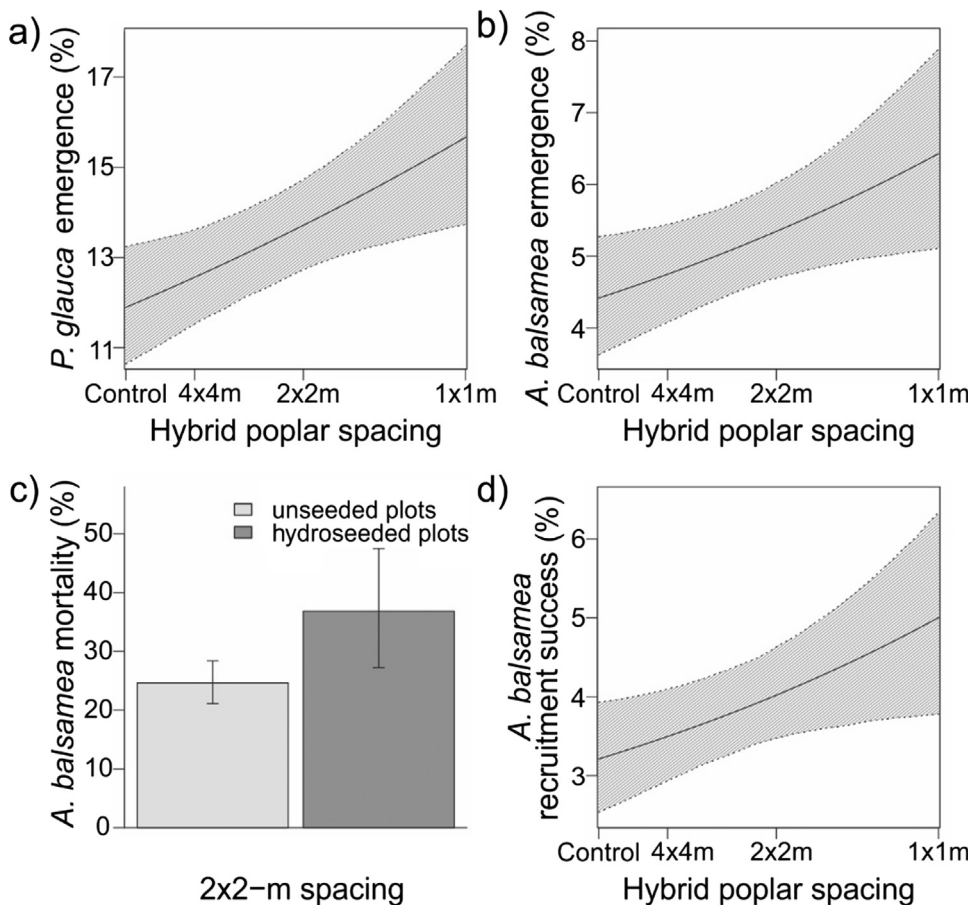


Fig. 5. Influence of experimental factors on seedling responses of *Picea glauca* and *Abies balsamea* during the 3rd growing season of the plantations. Results summarize (a) *P. glauca* and (b) *A. balsamea* summer emergence rates in relation to hybrid poplar spacing treatment, (c) *A. balsamea* summer mortality rates in relation to hydroseeded cover treatment and (d) recruitment success rates for *A. balsamea* first-year seedlings along the poplar gradient. Fitted values are presented with a 90% CI.

significant associations between the micro-environmental variables and *P. glauca* or *A. balsamea* seedling mortality.

4. Discussion

4.1. Nurse tree effect on the understory micro-environment

Young plantations of fast-growing trees on waste rock slopes rapidly exerted control over the understory conditions, in accordance with our first hypothesis. The micro-environment-modifying capacity of the nurse tree cover was observed for all understory parameters studied. However, the relationships between tree density and environmental conditions were often quadratic, indicating concurrent underlying mechanisms. The development of an overstory structure quickly reduced light availability in the young plantation understories, thereby driving changes in other micro-environment attributes. Stronger shading effect associated with denser hybrid poplar spacings likely

contributed to decreasing herbaceous biomass –largely attributable to shade-intolerant species– along the density gradient. Light is often the most limiting resource affecting understory plant establishment and growth (Strengbom et al., 2004; Hart and Chen, 2006). Conversely, light conditions can be modified by the understory vegetation, which in turn becomes a structural layer affecting the micro-environment (Burton and Bazzaz, 1991; Hart and Chen, 2006). Increased overstory light transmission is therefore frequently counterbalanced by increased light interception from the understory vegetation layer (Constabel and Liefers, 1996; Messier et al., 1998). In the plantations, the stronger structuring role played by the herbaceous cover under the widest spacing level and the control treatment without nurse tree was noticeable not only for available light at ground level, but also for other micro-environmental parameters, such as soil temperature and leaf litter cover. In other words, increased herbaceous biomass translated into a stronger contribution to light and heat interception and to litterfall.

The influence of a tree cover on surface soil moisture is known to be

Table 4

Correlations between naturally established *Salicaceae* (*Populus* and *Salix*) seedling emergence or mortality and micro-environmental variables. Results are based on the average value of each variable for the highest emergence (late June to late July 2014) and mortality (mid-July to mid-August 2015) periods. P-values are shown in bold when under the Bonferroni-adjusted significance threshold ($p = 0.10/5 = 0.02$). Confidence intervals (CI = $100 - 0.02 = 98\%$) excluding zero indicate a significant linear relationship at 2% probability of error.

	Emergence (2014)			Mortality (2015)		
	Coefficient	CI	P-value	Coefficient	CI	P-value
Herbaceous biomass	−0.538	−0.863; 0.099	0.047	0.403	−0.240; 0.800	0.136
Soil moisture	0.297	−0.350; 0.751	0.283	0.014	−0.596; 0.614	0.963
Available light	0.185	−0.170; 0.497	0.225	−0.399	−0.672; −0.029	0.013
Soil temperature	0.164	−0.191; 0.481	0.281	−0.075	−0.437; 0.308	0.655
Leaf litter cover	−0.388	−0.794; 0.256	0.153	−0.138	−0.670; 0.488	0.625

Table 5

Correlations between *Picea glauca* and *Abies balsamea* seedling emergence or mortality and micro-environmental variables. Results are based on average value of each variable for the highest emergence (mid-May to mid-June) and mortality (mid-July to mid-August) periods in 2015. P-values are shown in bold when under the Bonferroni-adjusted significance threshold ($p = 0.10/5 = 0.02$). Confidence intervals (CI = $100 - 0.02 = 98\%$) excluding zero (shown in bold) indicate a significant linear relationship at 2% probability of error.

	Emergence			Mortality		
	Coefficient	CI	P-value	Coefficient	CI	P-value
<i>Picea glauca</i>						
Herbaceous biomass	NA	NA	NA	0.013	−0.577; 0.595	0.962
Soil moisture	0.533	−0.077; 0.853	0.041	−0.182	−0.709; 0.475	0.533
Available light	−0.054	−0.391; 0.296	0.725	−0.092	−0.423; 0.261	0.549
Soil temperature	−0.248	−0.546; 0.105	0.100	0.158	−0.197; 0.477	0.299
Leaf litter cover	0.230	−0.124; 0.532	0.129	−0.490	−0.836; 0.135	0.064
<i>Abies balsamea</i>						
Herbaceous biomass	NA	NA	NA	0.440	−0.197; 0.816	0.101
Soil moisture	0.500	−0.122; 0.840	0.058	0.198	−0.463; 0.717	0.498
Available light	0.016	−0.330; 0.359	0.914	−0.018	−0.360; 0.329	0.908
Soil temperature	−0.144	−0.465; 0.211	0.345	−0.005	−0.350; 0.340	0.972
Leaf litter cover	−0.189	−0.501; 0.166	0.214	0.118	−0.503; 0.658	0.676

generally positive because of a reduced soil-to-air vapour pressure gradient and decreased wind movement, resulting in less evaporation from the soil surface (Geiger, 1965; Burton and Bazzaz, 1991; Man and Lieffers, 1999). Nonetheless, a significant decrease in moisture content was recorded during the drier second half of summer 2015 under the 1×1 -m cover compared to the 2×2 -m cover, while the relationship between poplar density and moisture content switched from positively linear to quadratic. Competition for water on drought-prone sites like waste rock slopes may thus negate the facilitative effect of the nurse shelter at high densities (Dordel, 2009). Moreover, fast-growing nurse trees may deplete soil water resources more rapidly than slower-growing species (Govindarajan et al., 1996; McIntyre et al., 1997). Rosenberg et al. (1983) noted that a fast-growing canopy can reduce soil moisture availability when transpiration rates exceed evaporation rates from bare soil. Thus, the trend observed in the plantations –that already started to appear in mid-summer 2014– likely reflected an offset of the beneficial tree cover shading effect by a transpiration effect for the highest planting density.

4.2. Nurse tree effect on forest tree seedlings

Our results partly support the hypothesis that planting of fast-growing nurse trees would benefit forest tree seedling performances (Lugo, 1997; Parrotta et al., 1997) compared to bare soil, but indicate that facilitation and competition were driven by differential response patterns among species. The densest cover treatment (1×1 m) promoted the highest emergence rates for *P. glauca* and *A. balsamea* seedlings. Sheltered conditions also improved the emergence of *Salicaceae* seedlings, while higher emergence rates were observed under the 2×2 -m spacing treatment. The emergence patterns observed may compare with data collected in forested environments. For example, McLaren and Janke (1996) found *A. balsamea* emergence to increase as a function of tree cover in *Populus tremuloides*-dominated stands. Similarly, Calogeropoulos et al. (2004) observed greater *A. balsamea* emergence in partial cuts than in clearcut treatments in their study on the effects of harvest intensity. Sheltered sites were also noted to produce more germinants than open sites for earlier-successional tree species (Burns and Barbara, 1990). Nonetheless, *Salicaceae* seedling mortality rates increased along the hybrid poplar density gradient during the final year of sampling (2015), likely pointing to an increasing exclusion of these species in the plantations, typical colonizers of more open habitats (Burns and Barbara, 1990; Peterson and Peterson, 1992).

The light-water model proposed by Holmgren et al. (1997) suggests that facilitation occurs when the improvement of plant-water

relationships under the canopy exceeds the costs of decreased light availability, hence hinting at the potential of nurse plantations on drought-prone slopes, especially for shade-tolerant species. Our correlation analysis using a conservative significance threshold did not allow to clearly pinpoint the determining micro-environmental variables for seedling emergence. Nonetheless, the highest emergence rates for both *A. balsamea* and *P. glauca* coincided with the highest soil moisture contents found during the peak emergence period (mid-May to mid-June 2015), while moisture levels increased with plantation density. Greater emergence for *Salicaceae* species also occurred under the spacing treatment associated with the highest soil moisture content values during the 2014 emergence period, i.e. the 2×2 -m spacing treatment. If soil moisture is known to be the most limiting factor for the germination of boreal tree species (Greene et al., 1999), light availability emerged as a driving factor of *Salicaceae* species abundance as soon as during the third growing season of the plantations. Decrease in understory light already became limiting for the survival of these shade-intolerant species, whose germinants lacking endosperm are more dependent on rapidly establishing photosynthetic leaf area to support development (Karrenberg et al., 2002).

Results for *Salicaceae* species showed that facilitation and competition mechanisms not only differ among plantation densities, but also change as plantations continue to develop and as seedlings grow. Pickett et al. (1987) remarked that early facilitation of a colonizer by a nurse plant often gives way to inhibition as the colonizer matures. Competition for light under denser cover started to outweigh facilitation and exerted a detrimental effect for *Salicaceae* seedlings in 2015. In this context, later-successional species such as *A. balsamea*, more sensitive to soil moisture than light limitations (Lieffers and Stadt, 1994; Landhäusser and Lieffers, 2001), may be better equipped to take advantage of opportunities in the understory. Nonetheless, the decrease in moisture content recorded at the end of the survey under the densest hybrid poplar spacing treatment suggests that their influence on soil water resources may become increasingly competitive over the next growing seasons, and compromise their nursing potential in the longer term.

4.3. Hydroseeded cover effect on forest tree seedlings

In accordance with our third hypothesis, the competitive effect exerted by the hydroseeded herbaceous cover limited seedling emergence and increased seedling mortality, although not all forest tree species were influenced in the same way. Dense herbaceous covers are known to reduce seedbed receptivity, especially for small-seeded species like poplars and willows, characterized by smaller radicles with limited

ability to reach soil resources (Greene and Johnson, 1998; Hesketh et al., 2009). Although we could only document the detrimental effect of the hydroseeded cover treatment on pioneer *Salicaceae* seedling emergence, we do not conclude that the hydroseeded cover would not affect the emergence of larger-seeded *P. glauca* and *A. balsamea* originating from seed rain. Our sowing method with direct positioning of seeds onto the mineral soil may have inflated emergence occurrences. Previous studies noted for instance that abundant herbaceous vegetation inhibits spruce regeneration: dense growth may intercept seeds before they reach the ground (Eis, 1981), while germinants that do reach the soil are more easily shaded out, smothered by dead foliage or outcompeted for soil water (Coates et al., 1994). Increase in mortality rates in hydroseeded plots were, however, only registered for *A. balsamea* seedlings, which may be somewhat surprising given that fir seeds are four times larger than white spruce seeds and usually lead to more resistant seedlings able to develop their root system more rapidly (Bakuzis and Hansen, 1965). Nonetheless, this result may reflect the difficulty for slower-growing seedlings to compete against aggressive herbaceous species in more open environments.

The understory vegetation can both act as a driver of tree succession (Hart and Chen, 2006) and be determined by overstory structure (De Grandpré et al., 1993). The negative relationship between hybrid poplar cover and herbaceous biomass monitored in the young plantations suggests that the increasingly dense tree cover will eventually shade out the light-demanding (naturally established or hydroseeded) ground cover vegetation. Therefore, the competitive effect of the hydroseeded cover on seedlings may weaken as the plantations mature.

4.4. Forest-like recruitment rates?

It was suggested that the rapid redevelopment of a tree cover could accelerate the return of conditions beneficial to the regeneration of forest tree seedlings (Carnevale and Montagnini, 2002; Boothroyd-Roberts et al., 2013). Under the densest tree cover treatment, recruitment success rate for *A. balsamea* first-year seedlings averaged 5%. This compares to results (4–4.5%) found in studies realized in aspen (*Populus tremuloides* Michaux)-dominated stands for seedlings less than two years old (McLaren and Janke, 1996; Calogeropoulos et al., 2004). Regardless of tree cover treatment, recruitment success rate for *P. glauca* was higher (~10%) than for *A. balsamea*, and surprisingly high compared to studies in forested environment (2–4%) (Purdy et al., 2002; Simard et al., 2003; Calogeropoulos et al., 2004; Wang and Kembell, 2005). Although success rates of first-year seedlings may be promising, our sowing method could again limit the interpretation of our results—especially for smaller-seeded *P. glauca* seedlings, more dependent on seedbed receptivity—, since seed positioning in tiny furrows may have positively biased emergence rates. Concordantly, success rates in control plots without nurse trees remained relatively high for both *A. balsamea* (> 3%) and *P. glauca* (~9%).

Results from the natural colonization survey showed that *A. balsamea* (2%) and *P. glauca* (10%) seedlings accounted for a small proportion of colonizing seedling abundance after three growing seasons. This translates into a rather marginal density of 0.02 and 0.10 seedlings per m² respectively for each species. While context-dependent and based on a limited period of time, these results may suggest that seed availability for species most likely to benefit from the understory conditions could be unreliable on uphill rock slopes. White spruce seeds are

known to disperse up to 300 m (Sims et al., 1990), but as low as 7, 4 and 0.1% of them generally reach 50, 100 and 200 m (Nienstaedt and Zasada, 1990). The distance is even smaller for the heavier fir seeds, many of which fall with their scales near the parent tree (Sims et al., 1990). Consequently, since successful seedling recruitment requires sufficient availability of seeds, management practices may have to include seeding work depending on the vicinity of seed bearers and site topography.

5. Conclusion

Natural processes such as succession provide a framework for the implementation of sustainable restoration practices. Plantations may act as accelerators of natural succession through rapid canopy re-establishment. Compared to the traditional anti-erosion treatment, results in the young plantations of hybrid poplars suggest that fast-growing plantations rapidly provide opportunities for the early establishment of later-successional boreal tree seedlings in waste rock slopes. This novel assemblage (Hobbs et al., 2006) relying on a semi-exotic tree species accelerated canopy redevelopment and already exerted a structural influence on the forest tree recruitment conditions in the first years after planting. Nonetheless, soil moisture monitoring results may imply that influence on soil water resources will become increasingly competitive as fast-growing hybrid poplars continue to develop.

Waste rock pile topography, characterized by erodible slopes, increases the importance of rapid tree establishment. Hybrid poplars have the advantage of growing more quickly than other nurse tree species available for reclamation (Guy and Bateman, 1989). However, high soil water consumption may compromise longer-term nursing potential of hybrid poplars in waste rock slopes where competition for limited resources, such as water, may be severe.

Declaration of interest

None for Hugo Bouchard, Marie Guittonny or Suzanne Brais.

Role of the funding bodies

This study was funded by the Natural Sciences and Engineering Research Council of Canada (NSERC), the Fonds de recherche du Québec – Nature et technologies (FRQNT) and the Canadian Malartic Partnership. The industrial partner (Canadian Malartic Partnership) was consulted during the preliminary study design phase in order to improve the applicability of the results. Nonetheless, funding bodies had no involvement in the collection, analysis and interpretation of data, in the writing of the manuscript, or in the decision to submit the manuscript for publication.

Acknowledgements

The authors are indebted to the research technician and undergraduate students who have contributed to the collection of field data, namely Avril Jobin, Yoan Goudreau, Charles Frenette-Vallières, Sara Gervais, Paulo-Bernardo Neves-E-Castro, Antoine Beaulieu, Maguelonne Rajot and Liam Boivin. We are also grateful to Brian Harvey, Guy LaRocque, Yves Bergeron and Jean-François Boucher for providing valuable comments.

Appendix A

List of herbaceous species inventoried during the 2nd and 3rd growing seasons in the hybrid poplar plantations

Common name	Scientific name
Alsike clover	<i>Trifolium hybridum</i> Linnaeus
Bull thistle	<i>Cirsium vulgare</i> (Savi) Tenore
Canada goldenrod	<i>Solidago canadensis</i> Linnaeus
Canada thistle	<i>Cirsium arvense</i> (Linnaeus) Scopoli
Coltsfoot	<i>Tussilago farfara</i> Linnaeus
Common dandelion	<i>Taraxacum officinale</i> F.H. Wiggers
Common Kochia	<i>Bassia scoparia</i> (Linnaeus) Voss
Common lamb's-quarters	<i>Chenopodium album</i> Linnaeus
Common mullein	<i>Verbascum thapsus</i> Linnaeus
Common plantain	<i>Plantago major</i> Linnaeus
Common timothy	<i>Phleum pratense</i> Linnaeus
Common yarrow	<i>Achillea millefolium</i> Linnaeus
Curled dock	<i>Rumex crispus</i> Linnaeus
Field sow-thistle	<i>Sonchus arvensis</i> Linnaeus subsp. <i>arvensis</i>
Fireweed	<i>Chamaenerion angustifolium</i> (Linnaeus) Scopoli subsp. <i>angustifolium</i>
Garden bird's-foot trefoil	<i>Lotus corniculatus</i> Linnaeus
Grass-leaved goldenrod	<i>Euthamia graminifolia</i> (Linnaeus) Nuttall
Heart-leaved aster	<i>Symphyotrichum cordifolium</i> (Linnaeus) G.L. Nesom
Horsetail sp.	<i>Equisetum</i> sp.
Large-leaved aster	<i>Eurybia macrophylla</i> (Linnaeus) Cassini
Narrow-leaved blue-eyed-grass	<i>Sisyrinchium angustifolium</i> Miller
Orange hawkweed	<i>Pilosella aurantiaca</i> (Linnaeus) F.W. Schultz & Schultz Bipontinus
Oxeye daisy	<i>Leucanthemum vulgare</i> Lamarck
Red clover	<i>Trifolium pratense</i> Linnaeus
Red fescue	<i>Festuca rubra</i> Linnaeus subsp. <i>rubra</i>
Red raspberry	<i>Rubus idaeus</i> Linnaeus
Reed canarygrass	<i>Phalaris arundinacea</i> Linnaeus
Rough cinquefoil	<i>Potentilla norvegica</i> Linnaeus
Rough-stemmed goldenrod	<i>Solidago rugosa</i> Miller
Sedge sp.	<i>Carex</i> sp.
Smartweed sp.	<i>Persicaria</i> sp.
Tufted vetch	<i>Vicia cracca</i> Linnaeus
White clover	<i>Trifolium repens</i> Linnaeus
White sweet-clover	<i>Melilotus albus</i> Medikus
Wild strawberry	<i>Fragaria virginiana</i> Miller
Yellow sweet-clover	<i>Melilotus officinalis</i> (Linnaeus) Lamarck

Appendix B

List of naturally established woody (tree and arborescent shrub) species inventoried during the 2nd and 3rd growing seasons in the hybrid poplar plantations

Common name	Scientific name
American mountain-ash	<i>Sorbus americana</i> Marshall
Balsam fir	<i>Abies balsamea</i> (Linnaeus) Miller
Balsam poplar	<i>Populus balsamifera</i> Linnaeus
Bebb's willow	<i>Salix bebbiana</i> Sargent
Black spruce	<i>Picea mariana</i> (Miller) BSP
Paper birch	<i>Betula papyrifera</i> Marshall
Pin cherry	<i>Prunus pensylvanica</i> Linnaeus f.
Prairie willow	<i>Salix humilis</i> Marshall
Pussy willow	<i>Salix discolor</i> Muhlenberg
Red-osier dogwood	<i>Cornus sericea</i> Linnaeus
Trembling aspen	<i>Populus tremuloides</i> Michaux

References

- Abe, K., Ziemer, R.R., 1991. Effect of tree roots on a shear zone: modeling reinforced shear stress. *Can. J. For. Res.* 21, 1012–1019.
- Agriculture and Agri-Food Canada, 2014. Length of Growing Season in Quebec, Baseline: 1971–2000. < <http://www.agr.gc.ca/eng/science-and-innovation/agricultural-practices/climate/future-outlook/climate-change-scenarios/length-of-growing-season-in-quebec/?id=1363104198111> > (retrieved 16.01.11).
- Andersen, C., Bussler, B., Chaney, W., Pope, P., Byrnes, W., 1989. Concurrent establishment of ground cover and hardwood trees on reclaimed mined land and unmined reference sites. *For. Ecol. Manage.* 28, 81–99.
- Ashby, W.C., 1997. Soil ripping and herbicides enhance tree and shrub restoration on stripmines. *Restor. Ecol.* 5, 169–177.
- Aubuchon, E.A., 2010. Soil moisture profiles and root growth of hardwood trees planted in different groundcovers on the steep slopes of reclaimed mine sites. M.S. Thesis, University of Tennessee, Knoxville, TN.
- Bakuzis, E., Hansen, H., 1965. Balsam fir, *Abies balsamea* (Linnaeus) Miller: A Monographic Review. The University of Minnesota Press, Minneapolis, Minn.
- Bates, D., Maechler, M., Bolker, B., Walker, S., 2015. lme4: Linear mixed-effects models using Eigen and S4. R package version 1.1–10. < <https://cran.r-project.org> > (retrieved 15.07.11).
- Bertness, M.D., Callaway, R., 1994. Positive interactions in communities. *Trends Ecol. Evol.* 9, 191–193.
- Boothroyd-Roberts, K., Gagnon, D., Truax, B., 2013. Can hybrid poplar plantations accelerate the restoration of forest understory attributes on abandoned fields? *For. Ecol. Manage.* 287, 77–89.
- Brooks, B., 1990. Overview of the manual of methods used in the revegetation of reactive sulphide tailings basins. In: Colloque sur la réduction et le drainage des effluents acides générés par les activités minières, Val d'Or, Quebec, pp. 247–268.
- Burger, J.A., Zipper, C.E., 2012. How to restore forests on surface-mined land. Virginia Cooperative extension publication. Virginia Cooperative Extension Service Publication No. 460-123. Virginia Polytechnic and State University, Blacksburg, VA.
- Burns, R.M., Barbara, H., 1990. Silvics of North America: 1. conifers; 2. hardwoods. Agriculture Handbook 654. U.S. Department of Agriculture, Forest Service, Washington, DC.
- Burton, P.J., Bazzaz, F., 1991. Tree seedling emergence on interactive temperature and moisture gradients and in patches of old-field vegetation. *Am. J. Bot.* 78, 131–149.
- Calogeropoulos, C., Greene, D., Messier, C., Brais, S., 2004. The effects of harvest intensity and seedbed type on germination and cumulative survivorship of white spruce and balsam fir in northwestern Quebec. *Can. J. For. Res.* 34, 1467–1476.
- Carnevale, N.J., Montagnini, F., 2002. Facilitating regeneration of secondary forests with the use of mixed and pure plantations of indigenous tree species. *For. Ecol. Manage.* 163, 217–227.
- Casselman, C.N., Fox, T.R., Burger, J.A., Jones, A.T., Galbraith, J.M., 2006. Effects of silvicultural treatments on survival and growth of trees planted on reclaimed mine lands in the Appalachians. *For. Ecol. Manage.* 223, 403–414.
- Charron, I., Greene, D., 2002. Post-wildfire seedbeds and tree establishment in the southern mixedwood boreal forest. *Can. J. For. Res.* 32, 1607–1615.
- Clark Ashby, W., 1995. Hybrid *Populus* in reclamation. *Int. J. Surface Min. Reclam.* 9, 69–71.
- Coates, K.D., Haeussler, S., Lindeburgh, S., Pojar, R., Stock, A., 1994. Ecology and silviculture of interior spruce in British Columbia. Canada-British Columbia Partnership Agreement on Forest Resource Development: FRDA II, Report 220, Forestry Canada/British Columbia Ministry of Forestry, BC.
- Colas, F., Bettez, M., 2014. Contrôle du séchage des graines après leur stratification au Centre de semences forestières de Berthier : implantation d'une nouvelle méthode par pesées, Ressources naturelles Québec. < <http://www.mfp.gouv.qc.ca/publications/forets/connaissances/recherche/Colas-Fabienne/Avis-technique-SGRE-9.pdf> > (retrieved 14.03.15).
- Connell, J.H., Slatyer, R.O., 1977. Mechanisms of succession in natural communities and their role in community stability and organization. *Am. Natural.* 111, 1119–1144.
- Constabel, A., Lieffers, V., 1996. Seasonal patterns of light transmission through boreal mixedwood canopies. *Can. J. For. Res.* 26, 1008–1014.
- Czapowskyj, M.M., 1978. Hybrid poplar on two anthracite coal-mine spoils: 10-year results. Dept. of Agriculture, Forest Service, Northeastern Forest Experiment Station. Research Note. Broomall, PA.
- De Grandpré, L., Gagnon, D., Bergeron, Y., 1993. Changes in the understory of Canadian southern boreal forest after fire. *J. Veg. Sci.* 4, 803–810.
- De Keersmaecker, L., Martens, L., Verheyen, K., Hermans, M., De Schrijver, A., Lust, N., 2004. Impact of soil fertility and insolation on diversity of herbaceous woodland species colonizing afforestations in Muizen forest (Belgium). *For. Ecol. Manage.* 188, 291–304.
- Del Moral, R., Walker, L.R., 2007. Environmental Disasters, Natural Recovery and Human Responses. Cambridge University Press New York, New York, NY.
- Densmore, R.V., 2005. Succession on subalpine placer mine spoil: Effects of revegetation with *Alnus viridis*, Alaska, USA. Arctic Antarctic Alpine Res. 37, 297–303.
- Dordel, J., 2009. Effects of nurse tree species on growth environment and physiology of underplanted *Toona ciliata* Roemer in subtropical Argentinean plantations. Doctoral dissertation. University of British Columbia, Vancouver, BC.
- Douglas, G.B., McIvor, I.R., Potter, J.F., Foote, L.G., 2010. Root distribution of poplar at varying densities on pastoral hill country. *Plant and soil* 333, 147–161.
- Drake, L., 1986. Survival and growth of conservation shrubs and trees, with thin-cover reclamation on acid substrate, Iowa, USA. *Environ. Geochem. Health* 8, 62–67.
- Dutta, R.K., Agrawal, M., 2002. Effect of tree plantations on the soil characteristics and microbial activity of coal mine spoil land. *Trop. Ecol.* 43, 315–324.
- Eis, S., 1981. Effect of vegetative competition on regeneration of white spruce. *Can. J. For. Res.* 11, 1–8.
- Emerson, P., Skousen, J., Ziemkiewicz, P., 2009. Survival and growth of hardwoods in brown versus gray sandstone on a surface mine in West Virginia. *J. Environ. Qual.* 38, 1821–1829.
- Espigares, T., Moreno-de las Heras, M., Nicolau, J.M., 2011. Performance of vegetation in reclaimed slopes affected by soil erosion. *Restorat. Ecol.* 19, 35–44.
- Fields-Johnson, C., Zipper, C., Burger, J., Evans, D., 2012. Forest restoration on steep slopes after coal surface mining in Appalachian USA: soil grading and seeding effects. *For. Ecol. Manage.* 270, 126–134.
- Filcheva, E., Noustrova, M., Gentcheva-Kostadinova, S., Haigh, M., 2000. Organic accumulation and microbial action in surface coal-mine spoils, Pernik, Bulgaria. *Ecol. Eng.* 15, 1–15.
- Franklin, J.A., Zipper, C.E., Burger, J.A., Skousen, J.G., Jacobs, D.F., 2012. Influence of herbaceous ground cover on forest restoration of eastern US coal surface mines. *New Forest.* 43, 905–924.
- Frouz, J., Vobořilová, V., Janoušová, I., Kadochová, Š., Matějčík, L., 2015. Spontaneous establishment of late successional tree species English oak (*Quercus robur*) and European beech (*Fagus sylvatica*) at reclaimed alder plantation and unreclaimed post mining sites. *Ecol. Eng.* 77, 1–8.
- Geiger, R., 1965. The Climate Near The Ground. (Translation of the German, 4th ed.). Harvard University Press, Cambridge, MA.
- Geldenhuys, C.J., 1997. Native forest regeneration in pine and eucalypt plantations in Northern Province, South Africa. *For. Ecol. Manage.* 99, 101–115.
- Gelman, A., Carlin, J.B., Stern, H.S., Rubin, D.B., 2014. Bayesian Data Analysis. Chapman & Hall/CRC Boca Raton, FL, USA.
- Government of Canada, 2015. Canadian Climate Normals 1981–2010 Station Data. < http://climate.weather.gc.ca/climate_normals/results/1981_2010_e.html?stnID=6019&ang=f&province=QC&provSubmit=go&page=1&dCode=> > (retrieved 16.01.11).
- Government of Quebec, 2017. Land Protection and Rehabilitation Regulation < <http://legisquebec.gouv.qc.ca/fr/ShowDoc/cr/Q-2,%20R.%2037?langCont=en> > (retrieved 17.11.11).
- Govindarajan, M., Rao, M.R., Mathuva, M.N., Nair, P., 1996. Soil-water and root dynamics under hedgerow intercropping in semiarid Kenya. *Agron. J.* 88, 513–520.
- Greene, D., Johnson, E., 1998. Seed mass and early survivorship of tree species in upland clearings and shelterwoods. *Can. J. For. Res.* 28, 1307–1316.
- Greene, D.F., Zasada, J.C., Sirois, L., Kneeshaw, D., Morin, H., Charron, I., Simard, M.J., 1999. A review of the regeneration dynamics of North American boreal forest tree species. *Can. J. For. Res.-Revue Can. Recherche Forestiere* 29, 824–839.
- Guariguata, M.R., Rheingans, R., Montagnini, F., 1995. Early woody invasion under tree plantations in Costa Rica: implications for forest restoration. *Restorat. Ecol.* 3, 252–260.
- Guy, P.R., Bateman, J.C., 1989. Determining optimal initial stocking densities during mine reclamation. In: Proceedings of the Conference - Reclamation, a Global Perspective, vols 1 and 2, pp. 317–326.
- Harrell Jr, F.E., Dupont, M.C., 2006. The Hmisc Package. R package version, 3, 0–12. < <https://cran.r-project.org> > (retrieved 16.11.04).
- Hart, S.A., Chen, H.Y., 2006. Understory vegetation dynamics of North American boreal forests. *Crit. Rev. Plant Sci.* 25, 381–397.
- Helm, D., 1995. Native grass cultivars for multiple revegetation goals on a proposed mine site in southcentral Alaska. *Restor. Ecol.* 3, 111–122.
- Heskeith, M., Greene, D., Pounden, E., 2009. Early establishment of conifer recruits in the northern Rocky Mountains as a function of postfire duff depth. *Can. J. For. Res.* 39, 2059–2064.
- Hobbs, R.J., Arico, S., Aronson, J., Baron, J.S., Bridgewater, P., Cramer, V.A., Epstein, P.R., Ewel, J.J., Klink, C.A., Lugo, A.E., 2006. Novel ecosystems: theoretical and management aspects of the new ecological world order. *Global Ecol. Biogeogr.* 15, 1–7.
- Holl, K.D., 2002. Long-term vegetation recovery on reclaimed coal surface mines in the eastern USA. *J. Appl. Ecol.* 39, 960–970.
- Holmgren, M., Scheffer, M., Huston, M.A., 1997. The interplay of facilitation and competition in plant communities. *Ecology* 78, 1966–1975.
- Johnstone, J.F., Chapin III, F.S., 2006. Effects of soil burn severity on post-fire tree recruitment in boreal forest. *Ecosystems* 9, 14–31.
- Jonasson, S., 1983. The point intercept method for non-destructive estimation of biomass. *Phytocoenologia* 11, 385–388.
- Jonasson, S., 1988. Evaluation of the point intercept method for the estimation of plant biomass. *Oikos* 101–106.
- Karrenberg, S., Edwards, P., Kollmann, J., 2002. The life history of *Salicaceae* living in the active zone of floodplains. *Freshw. Biol.* 47, 733–748.
- Kost, D.A., Vimmerstedt, J.P., 1994. Ground cover and tree growth on calcareous mine-soils: Greater influence of soil surface than nitrogen rate or seed mix. In: International Land Reclamation and Mine Drainage Conference and Third International Conference on The Abatement of Acidic Drainage: Proceedings. Volume 3: Reclamation and revegetation-SP 06C-94, pp. 295–304.
- Landhäusser, S.M., Lieffers, V.J., 2001. Photosynthesis and carbon allocation of six boreal tree species grown in understory and open conditions. *Tree Physiol.* 21, 243–250.
- Landhäusser, S.M., Rodriguez-Alvarez, J., Marenholtz, E.H., Lieffers, V.J., 2012. Effect of stock type characteristics and time of planting on field performance of aspen (*Populus tremuloides* Michx.) seedlings on boreal reclamation sites. *New Forest.* 43, 679–693.
- Laplace, P.S., 1986. Memoir on the probability of the causes of events. *Statist. Sci.* 1, 364–378.
- Larchevêque, M., Desrochers, A., Larocque, G.R., 2011. Comparison of manure compost and mineral fertilizer for hybrid poplar plantation establishment on boreal heavy clay soils. *Ann. For. Sci.* 68, 849–860.

- Lieffers, V., Stadt, K., 1994. Growth of understory *Picea glauca*, *Calamagrostis canadensis*, and *Epilobium angustifolium* in relation to overstory light transmission. *Can. J. For. Res.* 24, 1193–1198.
- Lugo, A.E., 1997. The apparent paradox of reestablishing species richness on degraded lands with tree monocultures. *For. Ecol. Manage.* 99, 9–19.
- Man, R.Z., Lieffers, V.J., 1999. Effects of shelterwood and site preparation on microclimate and establishment of white spruce seedlings in a boreal mixedwood forest. *For. Chron.* 75, 837–844.
- McGill, D.W., Ford, V.L., McNeel, J.F., 2004. Early development of a species test established on surface mines thirty years post-reclamation. In: Proceedings of Joint Conference of 21st Annual Meeting of the American Society of Mining and Reclamation, 25th West Virginia Surface Mine Drainage Task Force Symposium, Morgantown, WV, pp. 1227–1238.
- McIntyre, B., Riha, S., Ong, C., 1997. Competition for water in a hedge-intercrop system. *Field Crops Res.* 52, 151–160.
- McLaren, B.E., Janke, R.A., 1996. Seedbed and canopy cover effects on balsam fir seedling establishment in Isle Royale National Park. *Can. J. For. Res.* 26, 782–793.
- MERN, 2003. Vegetation Zones and Bioclimatic Domains in Québec. < <http://www.mern.gouv.qc.ca/english/publications/forest/publications/zone-a.pdf> > (retrieved 16.01.11).
- Messier, C., Parent, S., Bergeron, Y., 1998. Effects of overstory and understory vegetation on the understory light environment in mixed boreal forests. *J. Veg. Sci.* 9, 511–520.
- Naem, S., 2006. Biodiversity and ecosystem functioning in restored ecosystems: extracting principles for a synthetic perspective. In: Falk, D.A. (Ed.), *Foundations of Restoration Ecology*. Island Press, Washington, DC, pp. 210–237.
- Nichols, P.W., Morris, E.C., Keith, D.A., 2010. Testing a facilitation model for ecosystem restoration: Does tree planting restore ground layer species in a grassy woodland? *Austral Ecol.* 35, 888–897.
- Nienstaedt, H., Zasada, J.C., 1990. White spruce. *Silvics of North. America* 1, 389–442.
- Otsamo, R., 2000. Secondary forest regeneration under fast-growing forest plantations on degraded *Imperata cylindrica* grasslands. *New Forest.* 19, 69–93.
- Paquette, A., Messier, C., Perinet, P., Cogliastro, A., 2008. Simulating light availability under different hybrid poplar clones in a mixed intensive plantation system. *Forest Sci.* 54, 481–489.
- Parrotta, J.A., 1995. Influence of overstory composition on understory colonization by native species in plantations on a degraded tropical site. *J. Veg. Sci.* 6, 627–636.
- Parrotta, J.A., Turnbull, J.W., Jones, N., 1997. Catalyzing native forest regeneration on degraded tropical lands. *For. Ecol. Manage.* 99, 1–7.
- Peterson, E., Peterson, N., 1992. Ecology, management, and use of aspen and balsam poplar in the prairie provinces. *Special Rep.* 1, 252.
- Pickett, S., Collins, S., Armesto, J., 1987. Models, mechanisms and pathways of succession. *Botanical Rev.* 53, 335–371.
- Pietrzykowski, M., 2010. Scots pine (*Pinus sylvestris* L.) ecosystem macronutrients budget on reclaimed mine sites—stand trees supply and stability. *Natural Sci.* 2, 590.
- Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D., 2015. Nlme: linear and nonlinear mixed effects models. R package version 3.1-122. < <https://cran.r-project.org> > (retrieved 15.07.11).
- Polster, D., 2011. Effective reclamation: understanding the ecology of recovery. Paper presented at the 2011 Mine Closure Conference and BC Technical and Research Committee on Reclamation, BC Mine Reclamation Symposium. Lake Louise, AB.
- Polster, D.F., 2010. Long term reclamation monitoring of vegetative covers at the Island Copper Mine, Port Hardy, BC.
- Powers, J.S., Haggard, J.P., Fisher, R.F., 1997. The effect of overstory composition on understory woody regeneration and species richness in 7-year-old plantations in Costa Rica. *For. Ecol. Manage.* 99, 43–54.
- Purdy, B.G., Macdonald, S.E., Dale, M.R., 2002. The regeneration niche of white spruce following fire in the mixedwood boreal forest. *Silva Fennica* 36, 289–306.
- R Core Team –version 3.2.3–, 2015. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. Version 3.2.3. < <https://www.R-project.org> > (retrieved 15.04.22).
- Rizsa, J., Franklin, J., Buckley, D., 2007. The influence of different ground cover treatments on the growth and survival of tree seedlings on remined sites in eastern Tennessee. *Barnhisel RI (ed)* 30, 2–7.
- Rosenberg, N.J., Blad, B.L., Verma, S.B., 1983. *Microclimate: The Biological Environment*, 2nd ed. John Wiley & Sons, New York, NY.
- Simard, M.-J., Bergeron, Y., Sirois, L., 2003. Substrate and litterfall effects on conifer seedling survivorship in southern boreal stands of Canada. *Can. J. For. Res.* 33, 672–681.
- Sims, R., Kershaw, H.M., Wickware, G., 1990. The autecology of major tree species in the north central region of Ontario. Forestry Canada, Great Lakes Forest Research Centre, Sault Ste. Marie, ONT.
- Singh, A., Singh, J., 2006. Experiments on ecological restoration of coal mine spoil using native trees in a dry tropical environment, India: a synthesis. *New Forest.* 31, 25–39.
- Singh, A., Zeng, D., 2008. Effects of indigenous woody plantations on total nutrients of mine spoil in Singrauli Coalfield, India. *J. For. Res.* 19, 199–203.
- Singh, A.N., Raghubanshi, A.S., Singh, J.S., 2004. Impact of native tree plantations on mine spoil in a dry tropical environment. *For. Ecol. Manage.* 187, 49–60.
- Sloan, J.L., Jacobs, D.F., 2013. Fertilization at planting influences seedling growth and vegetative competition on a post-mining boreal reclamation site. *New Forest.* 44, 687–701.
- Soil Classification Working Group. 1998. The Canadian System of Soil Classification, 3rd ed. Agriculture and Agri-Food Canada. NRC Research press, Ottawa, ONT.
- Strengbom, J., Näsholm, T., Ericson, L., 2004. Light, not nitrogen, limits growth of the grass *Deschampsia flexuosa* in boreal forests. *Can. J. Botany* 82, 430–435.
- Torbert, J.L., Burger, J.A., 1994. Influence of grading intensity on ground cover establishment, erosion, and tree establishment on steep slopes. In: *International Land Reclamation and Mine Drainage Conference and Third International Conference on the Abatement of Acidic Drainage: Proceedings. Volume 3: Reclamation and Revegetation–SP 06C-94*, Pittsburgh, Pennsylvania, pp 226–231.
- Trindade, D.F.V., Coelho, G.C., 2012. Woody species recruitment under monospecific plantations of pioneer trees - facilitation or inhibition? *iForest – Biogeosci. Forestry* 5, 1–5.
- Walker, L.R., del Moral, R., 2009. Lessons from primary succession for restoration of severely damaged habitats. *Appl. Veget. Sci.* 12, 55–67.
- Wang, G.G., Kembell, K.J., 2005. Balsam fir and white spruce seedling recruitment in response to understory release, seedbed type, and litter exclusion in trembling aspen stands. *Can. J. For. Res.* 35, 667–673.
- Wilkinson, A., 1999. Poplars and willows for soil erosion control in New Zealand. *Biomass Bioenergy* 16, 263–274.
- Young, T., Petersen, D., Clary, J., 2005. The ecology of restoration: historical links, emerging issues and unexplored realms. *Ecol. Lett.* 8, 662–673.